Discrete Mathematics and Topics in Applied Discrete Mathematics

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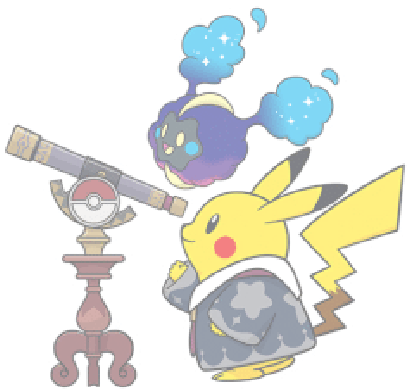


1 Counting

1.1 Permutation and Combination

Theorem. If a task can be performed in m ways or n ways, and the two sets of ways are disjoint, then the task can be performed in

$$m + n$$

ways.

Theorem. If a task consists of k successive independent steps with $n_1, n_2, \ldots, n_k$ choices respectively, then the task can be performed in

$$n_1 n_2 \cdots n_k$$

ways.

Definition. (Permutation) A permutation is an arrangement of objects in a specific order. If we have n objects, the number of ways to arrange r of them is given by the formula:

$$P(n, r) = \frac{n!}{(n-r)!}$$

where $n!$ (n factorial) is the product of all positive integers up to n, and represents the number of ways to arrange all n objects.

Proof. We are selecting and arranging r objects from a set of n distinct objects. Here's the reasoning:

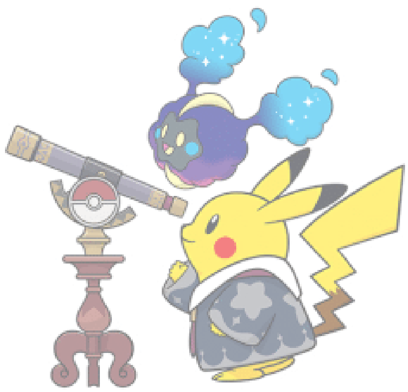
- For the first object, there are n choices.
- For the second object, there are $n - 1$ remaining choices.
- For the third object, there are $n - 2$ remaining choices, and so on.

Hence, the total number of ways to arrange r objects from n distinct objects is the product:

$$n \times (n-1) \times (n-2) \times \cdots \times (n-r+1)$$

This product is equivalent to

$$\frac{n!}{(n-r)!}$$



Definition. (Combination) A combination is a selection of objects without regard to order. The number of ways to choose r objects from n objects is given by the binomial coefficient:

$$C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Proof. Consider the total number of ways to choose r objects from n objects, assuming that the order matters (i.e., this is a permutation). The number of ways to arrange r objects from n is $P(n,r)$, which is given by

$$P(n,r) = \frac{n!}{(n-r)!}$$

However, in combinations, the order of selection does not matter. Therefore, we must divide by the number of ways to arrange the r objects chosen. The number of ways to arrange r objects is $r!$. Hence, the formula for combinations is

$$\binom{n}{r} = \frac{P(n,r)}{r!} = \frac{n!}{r!(n-r)!}$$

Example. How many ways can we choose a committee of 5 people from a group of 10 people, if 3 of the people must be selected, and the remaining 2 must be chosen from the remaining 7?

Solution. We divide the problem into two parts:

- First, we must choose the 3 people who are guaranteed to be on the committee. Since there are no restrictions, this can be done in $\binom{10}{3}$ ways.
- Next, we choose 2 more people from the remaining 7, which can be done in $\binom{7}{2}$ ways.

The total number of ways to form the committee is the product of the two combinations:

$$\binom{10}{3} \times \binom{7}{2} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} \times \frac{7 \times 6}{2 \times 1} = 2520$$

Example. Consider $n = 3$ pairs of parentheses. The number of valid parenthesis arrangements is C_3. We compute

$$C_3 = \frac{1}{3+1}\binom{6}{3} = \frac{1}{4} \times 20 = 5$$

The five valid arrangements of three pairs of parentheses are

$$1.\ ()()(), \quad 2.\ (())(), \quad 3.\ (()()), \quad 4.\ ((())), \quad 5.\ ()(())$$

Definition. A **combinatorial proof** is a proof of an identity that

- interprets both sides of an equation as counting the **same set**,
- uses logical counting arguments rather than algebraic manipulation.

Example.

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$$

Proof.

$$\binom{n}{r}$$

counts the number of ways to choose r objects from a set of n distinct objects.

$$\binom{n-1}{r} + \binom{n-1}{r-1}$$

counts the number of ways to choose r objects from the set, but in two cases:

- **Case 1:** We don't choose the n-th object. This leaves us with $n-1$ objects to choose from. The number of ways to choose r objects from these $n-1$ objects is $\binom{n-1}{r}$.
- **Case 2:** We choose the n-th object. In this case, we need to choose $r-1$ objects from the remaining $n-1$ objects. The number of ways to do this is $\binom{n-1}{r-1}$.

Example. (Binomial Theorem) For any integer $n \geq 0$,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

Proof. Consider the product

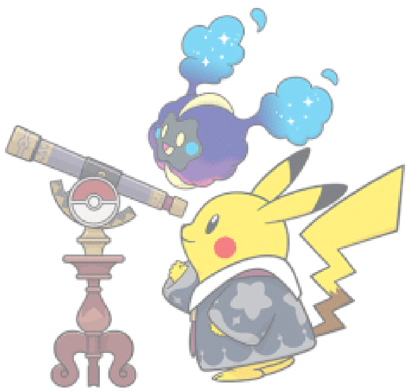
$$(a+b)^n = \underbrace{(a+b)(a+b)\cdots(a+b)}_{n \text{ factors}}$$

To obtain a term $a^{n-k}b^k$,

- choose b from exactly k of the n factors, and
- choose a from the remaining $n-k$ factor.

The number of ways to choose which k factors contribute a b is

$$\binom{n}{k}$$



Hence, the coefficient of $a^{n-k}b^k$ is $\binom{n}{k}$, proving the theorem.

Example. For any integer $n \geq 0$,

$$(x_1 + x_2 + \cdots + x_m)^n = \sum_{\substack{k_1+k_2+\cdots+k_m=n \\ k_i \geq 0}} \binom{n}{k_1, k_2, \ldots, k_m} x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m}$$

where the multinomial coefficient is defined as

$$\binom{n}{k_1, k_2, \ldots, k_m} = \frac{n!}{k_1! k_2! \cdots k_m!}$$

Proof. Consider

$$(x_1 + x_2 + \cdots + x_m)^n = \underbrace{(x_1 + x_2 + \cdots + x_m) \cdots (x_1 + x_2 + \cdots + x_m)}_{n \text{ factors}}$$

To obtain the term

$$x_1^{k_1} x_2^{k_2} \cdots x_m^{k_m}$$

we must

- choose x_1 from k_1 of the n factors,
- choose x_2 from k_2 of the remaining factors,
- $\cdots$
- choose x_m from the remaining k_m factors.

The number of such selections is

$$\frac{n!}{k_1! k_2! \cdots k_m!}$$

Hence, the coefficient is the multinomial coefficient, completing the proof.

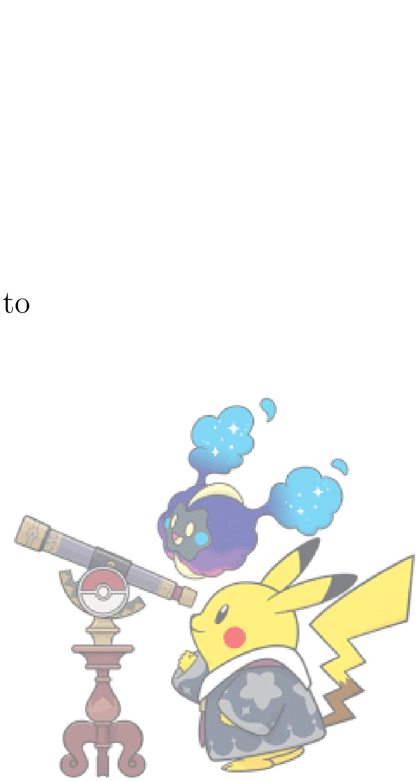
Example. (Stars and Bars) The number of nonnegative integer solutions to

$$x_1 + x_2 + \cdots + x_k = n$$

is

$$\binom{n+k-1}{k-1}$$

Proof. This is proved by interpreting solutions as arrangements of



- n identical stars, and
- $k-1$ dividers.

Theorem. (Combinations with Repetition) In some cases, repetitions are allowed when choosing objects. The number of ways to choose r objects from n distinct objects with repetition allowed is given by the following formula

$$\binom{n+r-1}{r}$$

This formula can be visualized as the number of ways to distribute r indistinguishable objects (choices) into n distinguishable boxes (objects). We are choosing r objects from n possible objects, and the formula accounts for the fact that repetitions are allowed.

Proof. We want to count the number of ways to choose r objects from n distinct objects where repetition is allowed. This can be thought of as finding the number of non-negative integer solutions to the equation:

$$x_1 + x_2 + \cdots + x_n = r$$

where x_i represents the number of times the i-th object is chosen, and each $x_i \geq 0$. By the stars and bars, the number of solutions to the equation

$$x_1 + x_2 + \cdots + x_k = r$$

where $x_i \geq 0$ for each i, is given by

$$\binom{r+k-1}{k-1}$$

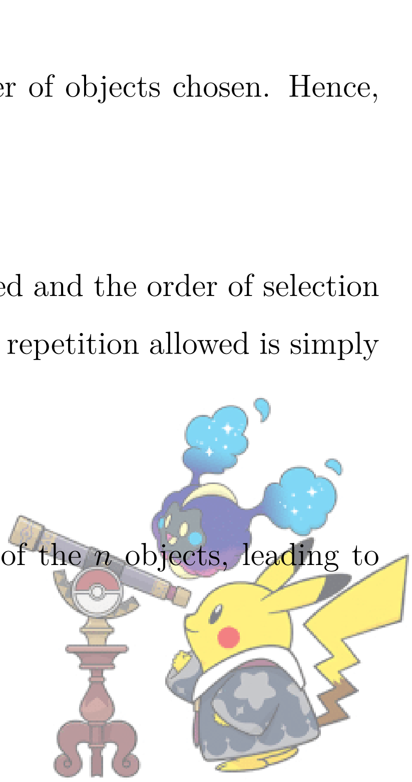
In our case, $k = n$ (the number of distinct objects) and r is the total number of objects chosen. Hence, the number of solutions is

$$\binom{r+n-1}{n-1} = \binom{n+r-1}{r}$$

Theorem. (Arrangements with Repetition) When repetitions are allowed and the order of selection matters, the number of ways to arrange r objects from n distinct objects with repetition allowed is simply

$$n^r$$

Since repetitions are allowed, each of the r positions can be filled with any of the n objects, leading to $n \times n \times \cdots \times n = n^r$ total arrangements.



1.2 Pigeonhole Principle

Theorem. If $n+1$ objects are placed into n boxes, then at least one box contains at least two objects.

Proof. Assume, for contradiction, that each box contains at most one object. Then the total number of objects is at most n, contradicting the fact that $n+1$ objects were placed. Therefore, at least one box must contain two or more objects.

Theorem. If N objects are distributed among k boxes, then at least one box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

Proof. Suppose every box contains at most $\left\lceil \frac{N}{k} \right\rceil - 1$ objects. Then the total number of objects is at most

$$k\left(\left\lceil \frac{N}{k} \right\rceil - 1\right) < N$$

which contradicts the assumption that there are N objects. Hence at least one box contains at least $\left\lceil \frac{N}{k} \right\rceil$ objects.

1.3 Inclusion-Exclusion Principle

For finite sets A, B, and C,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

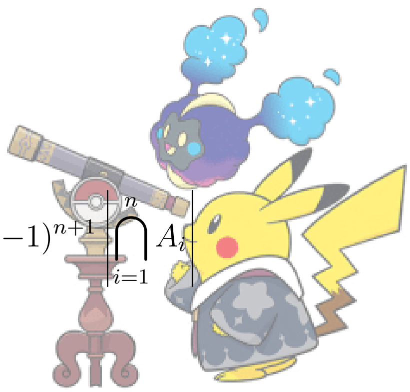
Proof.

- Each element is counted once for every set it belongs to in $|A| + |B| + |C|$.
- Elements belonging to exactly two sets are counted twice and subtracted once.
- Elements belonging to all three sets are counted three times, subtracted three times, and must be added back once.

Hence, each element of $A \cup B \cup C$ is counted exactly once.

Theorem. Let $A_1, A_2, \ldots, A_n$ be finite sets. Then

$$\left|\bigcup_{i=1}^{n} A_i\right| = \sum_{i} |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left|\bigcap_{i=1}^{n} A_i\right|$$



Proof. We proceed by induction on n, the number of sets. For $n = 2$, we have proven it. Assume the formula holds for some $k \geq 2$ sets:

$$\left|\bigcup_{i=1}^{k} A_i\right| = \sum_{\emptyset \neq J \subseteq \{1,\ldots,k\}} (-1)^{|J|-1} \left|\bigcap_{j \in J} A_j\right|$$

Consider the union of $k+1$ sets. We can group the first k sets together:

$$\begin{aligned}
\left|\bigcup_{i=1}^{k+1} A_i\right| &= \left|\left(\bigcup_{i=1}^{k} A_i\right) \cup A_{k+1}\right| \\
&= \left|\bigcup_{i=1}^{k} A_i\right| + |A_{k+1}| - \left|\left(\bigcup_{i=1}^{k} A_i\right) \cap A_{k+1}\right| \quad \text{(by base case)} \\
&= \left|\bigcup_{i=1}^{k} A_i\right| + |A_{k+1}| - \left|\bigcup_{i=1}^{k} (A_i \cap A_{k+1})\right| \quad \text{(by distributive law)} \\
&= \sum_{\emptyset \neq J \subseteq \{1,\ldots,k\}} (-1)^{|J|-1} \left|\bigcap_{j \in J} A_j\right| + \sum_{\emptyset \neq J \subseteq \{1,\ldots,k\}} (-1)^{|J|-1} \left|\bigcap_{j \in J} (A_j \cap A_{k+1})\right| \\
&\quad \text{(by inductive hypothesis to the first and third terms)}
\end{aligned}$$

The first sum accounts for all non-empty subsets of $\{1, \ldots, k+1\}$ that do not contain the index $k+1$. The second sum (with the negative sign distributed) accounts for all subsets that contain $k+1$ and at least one other index from $\{1, \ldots, k\}$. The $|A_{k+1}|$ term accounts for the singleton set $\{k+1\}$. Combining these yields the sum over all non-empty subsets $J \subseteq \{1, \ldots, k+1\}$:

$$\left|\bigcup_{i=1}^{k+1} A_i\right| = \sum_{\emptyset \neq J \subseteq \{1,\ldots,k+1\}} (-1)^{|J|-1} \left|\bigcap_{j \in J} A_j\right|$$

which completes the inductive step.

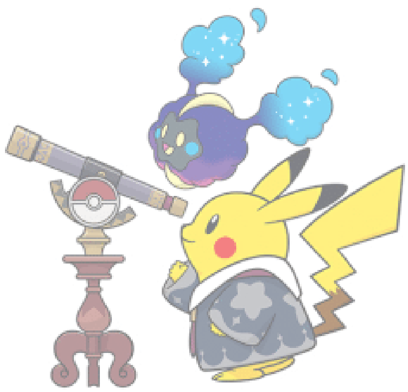
1.4 Recurrence Relation

Definition. A sequence $\{a_n\}$ is defined by

$$a_n = f(a_{n-1}, a_{n-2}, \ldots, a_{n-k})$$

together with **initial conditions**

$$a_0, a_1, \ldots, a_{k-1}$$



is called a recurrence relation. A **linear recurrence relation of order** k has the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}$$

where $c_1, \ldots, c_k$ are constants.

Assume a solution of the form

$$a_n = r^n$$

where $r \neq 0$ is a constant. Substituting into the recurrence gives

$$r^n \left(1 + c_1 r^{-1} + c_2 r^{-2} + \cdots + c_k r^{-k}\right) = 0$$

Dividing by r^{n-k} yields the **characteristic equation**:

$$r^k + c_1 r^{k-1} + c_2 r^{k-2} + \cdots + c_k = 0$$

If the characteristic equation has k distinct roots

$$r_1, r_2, \ldots, r_k$$

then the general solution is

$$a_n = C_1 r_1^n + C_2 r_2^n + \cdots + C_k r_k^n$$

where $C_1, \ldots, C_k$ are constants determined by initial conditions. If a root r has multiplicity m, then the corresponding part of the solution is

$$a_n = \left(C_0 + C_1 n + C_2 n^2 + \cdots + C_{m-1} n^{m-1}\right) r^n$$

Now consider

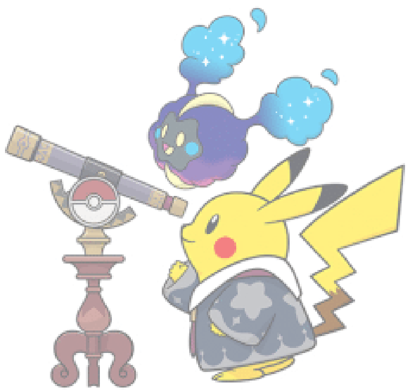
$$a_n + c_1 a_{n-1} + \cdots + c_k a_{n-k} = f(n)$$

The general solution is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

where

- $a_n^{(h)}$ solves the homogeneous equation,



- $a_n^{(p)}$ is a particular solution.

We first guess the form of $a_n^{(p)}$ based on $f(n)$.

$f(n)$	Trial particular solution
$A\alpha^n$	$B\alpha^n$
Polynomial in n of degree d	Polynomial of degree d

However, if the guessed form **overlaps** with the homogeneous solution, we must **modify** it by multiplying by powers of n. This is known as the **modification rule**.

1.5 Generating Function

Definition. (Generating Function) A **generating function** is a formal power series:

$$G(x) = \sum_{n=0}^{\infty} a_n x^n$$

where the coefficient a_n encodes a counting sequence.

Properties. Generating functions are linear, meaning if $G_1(x)$ and $G_2(x)$ are the generating functions of sequences $\{a_n\}$ and $\{b_n\}$, respectively, then the generating function of the sequence $\{a_n + b_n\}$ is the sum of the generating functions:

$$G(x) = G_1(x) + G_2(x)$$

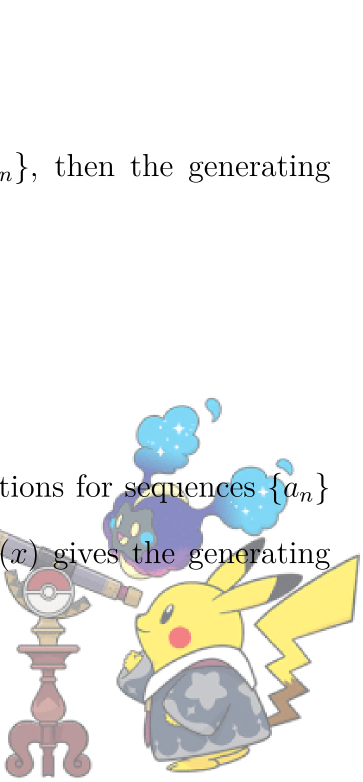
This property also holds for scalar multiples. For a constant c, we have generating function of $c \cdot \{a_n\}$ is given by

$$c \cdot G(x)$$

Properties. If $G(x) = \sum_{n=0}^{\infty} a_n x^n$ is the generating function of a sequence $\{a_n\}$, then the generating function for the sequence $\{a_{n+k}\}$ (shifted by k) is given by

$$\sum_{n=k}^{\infty} a_{n-k} x^n = x^k G(x)$$

Properties. If $G_1(x) = \sum_{n=0}^{\infty} a_n x^n$ and $G_2(x) = \sum_{n=0}^{\infty} b_n x^n$ are the generating functions for sequences $\{a_n\}$ and $\{b_n\}$, then the product of the two generating functions $G(x) = G_1(x) \cdot G_2(x)$ gives the generating



function of the convolution of the sequences:

$$G(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) x^n$$

Properties. If $G(x) = \sum_{n=0}^{\infty} a_n x^n$ is the generating function of the sequence $\{a_n\}$, then the generating function of the sequence $\{na_n\}$ is given by

$$\sum_{n=0}^{\infty} n a_n x^{n-1} = \frac{d}{dx} G(x)$$

Also, the generating function for the sequence $\{a_n\}$ raised to a power (say, square) can be computed by multiplying the generating function by itself.

Example. (Catalan Number) The Catalan numbers C_n count the number of ways to correctly arrange n pairs of parentheses. The formula for the n-th Catalan number is given by

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

Solution. A valid arrangement of parentheses means that in any prefix of the arrangement, the number of closing parentheses must not exceed the number of opening parentheses. This ensures that every closing parenthesis has a corresponding opening parenthesis.

We can describe the problem recursively. Let C_n represent the number of valid parenthesis arrangements with n pairs of parentheses. A valid arrangement starts with an opening parenthesis '(', which must eventually be matched with a closing parenthesis ')'. After the first '(' and its corresponding ')', the remaining part of the arrangement must also be a valid arrangement. Hence,

$$C_n = \sum_{k=0}^{n-1} C_k \cdot C_{n-1-k}$$

Here, C_k represents the number of valid arrangements for the first part, and C_{n-1-k} represents the number of valid arrangements for the second part. The base case is when $n = 0$, meaning there are no parentheses. There is exactly one valid arrangement:

$$C_0 = 1$$

Let the generating function $G(x)$ be defined as the power series:

$$G(x) = \sum_{n=0}^{\infty} C_n x^n$$

Substituting the recurrence relation into this generating function:

$$G(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n-1} C_k \cdot C_{n-1-k} \right) x^n$$

This can be split as:

$$G(x) = \sum_{n=0}^{\infty} C_0 x^n + \sum_{n=1}^{\infty} \left(\sum_{k=0}^{n-1} C_k \cdot C_{n-1-k} \right) x^n = 1 + \sum_{n=1}^{\infty} \left(\sum_{k=0}^{n-1} C_k \cdot C_{n-1-k} \right) x^n$$

The double sum can be recognized as the product of the generating function:

$$G(x) = 1 + xG(x)^2$$

Rearranging the equation:

$$G(x) - 1 = xG(x)^2$$

$$xG(x)^2 - G(x) + 1 = 0$$

Solving this quadratic equation for $G(x)$:

$$G(x) = \frac{1 - \sqrt{1 - 4x}}{2x}$$

Now, we need to extract the coefficient of x^n in the expansion of $G(x)$. The closed form of the Catalan number is given by

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

We can now expand $G(x)$ to find the coefficients corresponding to each C_n:

$$G(x) = \sum_{n=0}^{\infty} C_n x^n = \frac{1 - \sqrt{1 - 4x}}{2x}$$

Definition. (Stirling's Number) The Stirling numbers of the first kind, denoted $S(n, k)$, count the number of ways to arrange n elements in k disjoint cycles. These numbers are related to permutations.

The recurrence relation for the Stirling numbers of the first kind is given by

$$S(n,k) = S(n-1,k-1) + (n-1) \cdot S(n-1,k)$$

with the initial conditions:

$$S(0,0) = 1, \quad S(n,0) = 0 \text{ for } n > 0, \quad S(n,n) = 1$$

The Stirling numbers of the second kind, denoted $S(n,k)$, count the number of ways to partition a set of n elements into k non-empty subsets. These numbers are essential in combinatorics when studying partitions of sets. The recurrence relation for the Stirling numbers of the second kind is

$$S(n,k) = k \cdot S(n-1,k) + S(n-1,k-1)$$

with the initial conditions:

$$S(0,0) = 1, \quad S(n,0) = 0 \text{ for } n > 0, \quad S(n,n) = 1$$

2 Graph Theory

2.1 Introduction

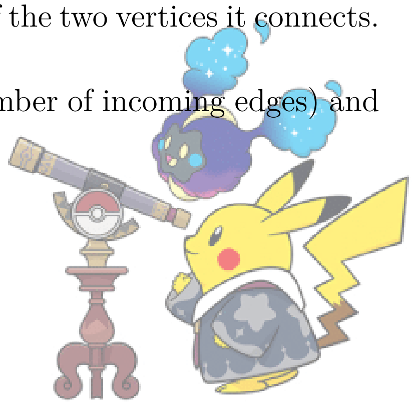
Definition. A graph is denoted as $G = (V,E)$, where

- V is the set of vertices (also called nodes).
- E is the set of edges (also called arcs) connecting pairs of vertices.

Definition. The **degree** of a vertex v in a graph is the number of edges incident to the vertex. This is denoted as $\deg(v)$.

- In an undirected graph, each edge contributes one to the degree of each of the two vertices it connects.
- In a directed graph, the degree is split into two types: *in-degree* (the number of incoming edges) and *out-degree* (the number of outgoing edges).

Definition. (Types of Graph)



- **Simple Graph:** A graph with no loops or multiple edges.
- **Directed Graph (Digraph):** A graph where the edges have a direction (i.e., each edge is an ordered pair of vertices).
- **Undirected Graph:** A graph where edges have no direction.
- **Weighted Graph:** A graph where each edge has a weight or cost associated with it.
- **Complete Graph:** A graph in which there is an edge between every pair of vertices.
- **Bipartite Graph:** A graph whose vertices can be divided into two disjoint sets such that no two vertices within the same set are adjacent.

Definition. (Isomorphism) Two graphs $G = (V_G, E_G)$ and $H = (V_H, E_H)$ are said to be **isomorphic** if there exists a one-to-one correspondence between their vertices and edges that preserves adjacency. Formally, there exists a bijection $f : V_G \to V_H$ such that for every edge $(u, v) \in E_G$, the corresponding edge $(f(u), f(v)) \in E_H$.

Definition. (Walk) A **walk** in a graph is a sequence of vertices and edges in which each edge is incident to its preceding and succeeding vertices. A **path** is a walk in which all vertices are distinct. A **trail** is a walk in which all edges are distinct.

Definition. (Cycle) A **cycle** (or circuit) is a path that starts and ends at the same vertex, with no repeated vertices except for the starting and ending vertex. A cycle is denoted as

$$v_1 \to v_2 \to \cdots \to v_n \to v_1$$

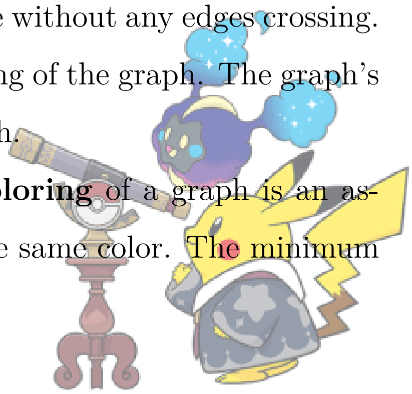
Definition. An **Eulerian circuit** in an undirected graph $G = (V, E)$ is a cycle that includes every edge of G exactly once. That is, the graph is traversed in such a way that each edge is visited once, and the starting vertex is the same as the ending vertex.

Definition. (Connected Graph) A graph is **connected** if there is a path between every pair of vertices in the graph. Otherwise, the graph is **disconnected**.

Definition. (Planar Graph) A graph is **planar** if it can be drawn on a plane without any edges crossing.

Definition. (Face) A **face** is a region bounded by edges in a planar embedding of the graph. The graph's faces include the outer face, which is the infinite region surrounding the graph.

Definition. (Vertex Coloring and Chromatic Number) A **vertex coloring** of a graph is an assignment of colors to the vertices such that no two adjacent vertices share the same color. The minimum



number of colors required to properly color a graph is called the **chromatic number** of the graph.

$$\chi(G) = \min\{k \mid G \text{ can be colored with } k \text{ colors}\}$$

Definition. (Matching) A **matching** is a set of edges such that no two edges share a common vertex. A **perfect matching** is a matching that covers all the vertices of the graph, i.e., every vertex is incident to exactly one edge in the matching.

Theorem. (Handshaking Lemma) In any undirected graph, the sum of the degrees of all vertices is twice the number of edges:

$$\sum_{v \in V} \deg(v) = 2|E|$$

Proof. Each edge in the graph contributes 1 to the degree of two vertices. Hence, counting the degree of each vertex for every edge gives twice the number of edges.

Theorem. (Euler's Polyhedron Formula for a Connected Planar Graph) For a planar and connected graph,

$$V - E + F = 2$$

where V is the number of vertices, E is the number of edges and F is the number of face.

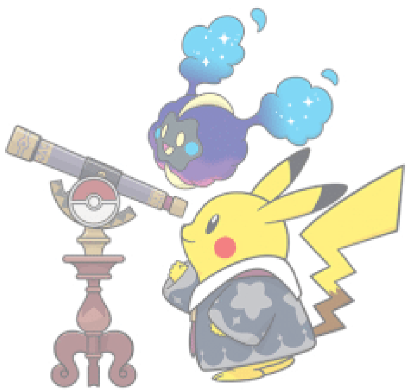
Remark. Beginners often forget to count the outer face. Even if a graph looks like a simple square on a piece of paper, it divides the paper into two regions: the inside of the square and the infinite space surrounding it. Both are counted as faces.

Proof. Apply induction on the number of faces (F) in the graph.

Base case: $F = 1$: A connected planar graph with only one face is, by definition, a tree. In any tree, the number of edges is always one less than the number of vertices: $E = V - 1$. Since there are no cycles, there is only the 1 outer face ($F = 1$). Then $V - E + F = 2$ holds.

Inductive steps: Assume the formula $V - E + F = 2$ is true for any connected planar graph with exactly k faces. Now, consider a connected planar graph G that has $k+1$ faces. As $F > 1$, the graph must contain at least one cycle. If it didn't, it would be a tree and only have one face. Remove an edge by picking an edge e that is part of a cycle. Let's call this new simplified graph G'. Since G' has k faces, our Inductive Hypothesis tells us it must satisfy the formula:

$$V' - E' + F' = 2$$



Now, substitute the values of G' in terms of the original graph G: $V' = V, \quad E' = E - 1, \quad F' = F - 1.$

Theorem. (Euler's Theorem for Eulerian Circuit) A connected graph has an Eulerian circuit if and only if every vertex has an even degree.

Proof. ($\Longrightarrow$) Assume that G has an Eulerian circuit. An Eulerian circuit is a cycle that visits every edge of the graph exactly once and returns to the starting vertex. In order for the circuit to exist, each vertex must have an even number of edges, so that when we enter a vertex, there is always a corresponding edge to leave the vertex. This ensures that we can traverse through the vertex twice—once entering and once exiting—without being left with an unmatched edge. Hence, if G has an Eulerian circuit, every vertex must have an even degree.

($\Longleftarrow$) Assume that every vertex in G has an even degree. We need to show that G has an Eulerian circuit. Note that since G is connected, there is a path between any two vertices. Start at any vertex (called v_0). Since the degree of each vertex is even, there will always be an even number of edges incident to each vertex, meaning that each time we enter a vertex, there will always be an unused edge to exit from it. Therefore, we can traverse the graph by following unused edges, and every vertex we visit will have an unused edge to continue the path.

As we follow the edges, we are guaranteed that no vertex will be left with an unpaired entry edge. Since the graph is connected and every vertex has an even degree, we can complete the traversal of all edges and return to the starting vertex. This forms an Eulerian circuit.

Theorem. (Kuratowski's Theorem) A graph is planar if and only if it does not contain a subgraph that is a subdivision of either K_5 (complete graph on 5 vertices) or $K_{3,3}$ (complete bipartite graph with parts of size 3).

Theorem. (Hall's Marriage Theorem) A bipartite graph $G = (X, Y, E)$ (with bipartite sets X and Y, and edge sets E) has a perfect matching if and only if for every subset $S \subseteq X$, the number of neighbors of S is at least as large as the size of S:

$$|N(S)| \geq |S| \quad \text{for all} \quad S \subseteq X$$

where $N(S)$ denotes the set of neighbors of S. Neighbors of a vertex are the vertices in the opposite set that are adjacent to the given vertex.

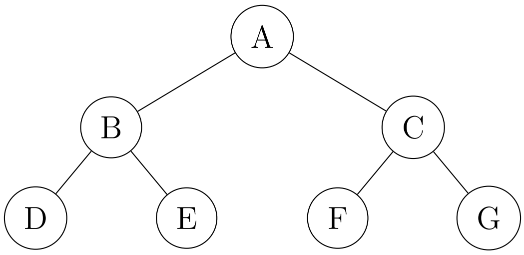
Theorem. (Turán's Theorem) The maximum number of edges in a graph on n vertices that does not contain a complete subgraph K_{r+1} is given by

$$T(n, r) = \left(1 - \frac{1}{r}\right)\frac{n^2}{2}$$

2.2 Tree

Definition. (Tree) A **tree** is a connected graph with no cycles.

Example.



Theorem. A graph with n vertices is a tree if and only if it has exactly $n-1$ edges and is connected.

Proof. ($\Longrightarrow$) We prove by induction on the number of vertices n that a tree with n vertices has exactly $n-1$ edges. A tree with one vertex has no edges. Since $1-1=0$, the statement holds. Assume that any tree with k vertices has exactly $k-1$ edges.

Let T be a tree with $k+1$ vertices. As T is a tree, it must have at least one **leaf** (a vertex of degree 1). Remove this leaf and the single edge incident to it. The resulting graph T' has k vertices. Because removing a leaf cannot disconnect the graph or create a cycle, T' is still a tree. By the induction hypothesis, T' has $k-1$ edges. Adding back the removed vertex and edge gives T exactly

$$(k-1)+1=k$$

edges. Since $k+1=n$, the number of edges is $n-1$. Hence, any tree with n vertices has exactly $n-1$ edges and is connected.

($\Longleftarrow$) Let G be a connected graph with n vertices and $n-1$ edges.

Suppose, for the sake of contradiction, that G contains a cycle. Removing any edge from this cycle does not disconnect the graph, since there is an alternative path along the cycle. After removing one such edge, the graph remains connected but now has $n-2$ edges.

This contradicts the fact that any connected graph on n vertices must have at least $n-1$ edges. Therefore, G contains no cycles. Since G is connected and acyclic, it is a tree.

Properties.

- A tree is minimally connected, meaning that the removal of any edge will disconnect the graph.
- A tree has exactly one path between any two vertices.

- A tree is a bipartite graph, meaning that its vertex set can be divided into two disjoint sets such that no two vertices within the same set are adjacent.
- Any connected graph with n vertices and $n-1$ edges is a tree.
- A tree with n vertices always has at least one leaf (a vertex with degree 1).
- Removing any edge from a tree results in a graph with exactly two connected components.
- Any graph with n vertices and $n-1$ edges that does not contain any cycles is a tree.

Proof.

- Let T be a tree with n vertices. By definition, a tree is connected and has no cycles. Assume for contradiction that we can remove an edge e from T and still keep the graph connected. If the graph remains connected, there would be another path between the two vertices that were previously connected by e, forming a cycle. This contradicts the assumption that T is acyclic. Therefore, removing any edge from a tree disconnects the graph, proving that a tree is minimally connected.
- Suppose there are two distinct paths between vertices u and v in a tree T. If there were two paths, one of the paths would contain a cycle, which contradicts the acyclicity of a tree. Therefore, there must be exactly one path between any two vertices in a tree.
- We can color the vertices of a tree using two colors, say A and B, in such a way that every edge connects a vertex from A to a vertex from B. Start by coloring an arbitrary vertex in color A. All adjacent vertices are colored with color B, and the process is repeated for all vertices. Since the tree has no cycles, we can always color the vertices without having two adjacent vertices share the same color. Thus, the tree is bipartite.
- Let G be a connected graph with n vertices and $n-1$ edges. Suppose, for contradiction, that G is not a tree. Then, G must contain a cycle. However, a cycle requires at least n edges (since there are n vertices in the cycle), which contradicts the assumption that G has only $n-1$ edges. Therefore, G must be acyclic and connected, which implies it is a tree.
- Suppose, for contradiction, that a tree has no leaves, i.e., every vertex has degree at least 2. In a tree with n vertices, the total degree is $2(n-1)$, since every edge contributes 2 to the total degree. If every vertex has degree at least 2, the total degree is at least $2n$, which is a contradiction. Therefore, there must be at least one vertex with degree 1, and thus a tree always has at least one leaf.

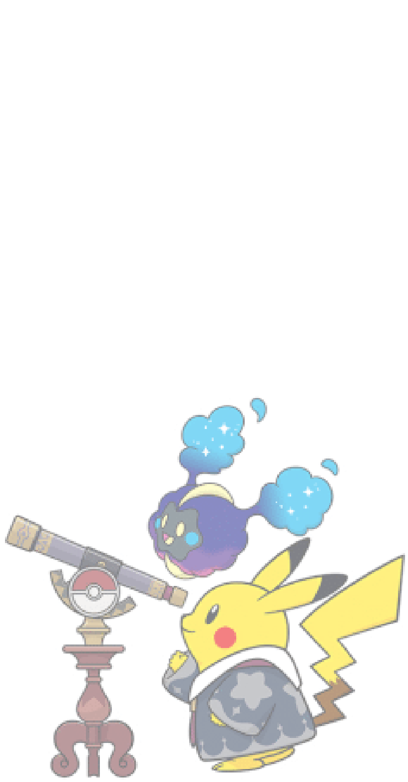
- Let T be a tree with n vertices and $n-1$ edges. If we remove an edge e from T, we are left with two subgraphs. Since T is connected, removing e must disconnect the graph into two components, and there is no path between any vertex in one component to any vertex in the other component. Therefore, the removal of any edge from a tree results in exactly two connected components.

- Let G be a graph with n vertices and $n-1$ edges that contains no cycles. By the previous proof, we know that a connected graph with n vertices and $n-1$ edges is a tree. Since G is acyclic, if it is also connected, it must be a tree. Therefore, any graph with n vertices and $n-1$ edges that does not contain any cycles is a tree.

2.3 Ramsey Number

Definition. (Ramsey Number) The Ramsey number $R(s,t)$ is the smallest number of vertices n such that any two-coloring of the edges of the complete graph on n vertices, K_n, will contain a monochromatic complete subgraph on k vertices. The followings are known Ramsey number:

s	t	$R(s,t)$
1	1	1
2	2	6
2	3	9
2	4	18
3	3	6
3	4	9
3	5	14
4	4	18
4	5	25
4	6	35
5	5	43
5	6	49
5	7	58
6	6	102
7	7	108
8	8	165
9	9	435



Properties.

- The Ramsey number is symmetric in its arguments:

$$R(m, n) = R(n, m)$$

- Increasing one of the parameters increases (or leaves unchanged) the Ramsey number:

$$R(m, n) \leq R(m + 1, n) \quad \text{and} \quad R(m, n) \leq R(m, n + 1)$$

  This is a consequence of the fact that it is harder to avoid a monochromatic complete subgraph as we increase the size of the desired subgraph.

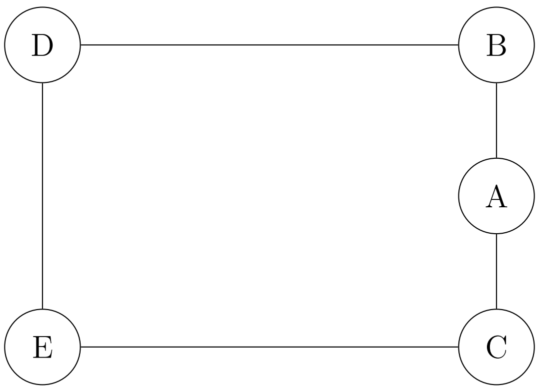
- For all $m \leq n$,

$$R(m, n) \leq R(m, m)$$

2.4 Five Color Theorem

Theorem. Any map can be colored using at most five colors, such that no two adjacent regions share the same color.

Remark. A map can be represented by a planar graph:



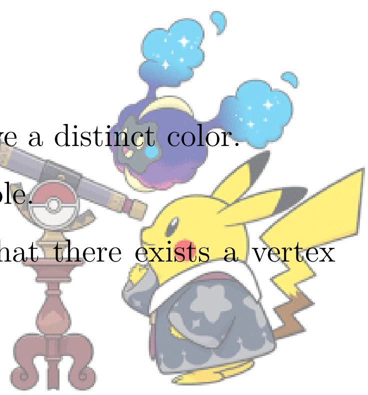
This map has five regions, and we need to assign colors to them such that no two adjacent regions share the same color.

Proof. We proceed by induction on the number of vertices n.

Base Case: For $n \leq 5$, the graph is clearly 5-colorable as each vertex can receive a distinct color.

Inductive Hypothesis: Assume all planar graphs with n vertices are 5-colorable.

Inductive Step: Let G be a planar graph with $n + 1$ vertices. It is known that there exists a vertex



$v \in G$ such that $\deg(v) \leq 5$.

Consider the graph $G' = G - v$. Since G' has n vertices, it is 5-colorable by the inductive hypothesis. Let the colors be $\{1, 2, 3, 4, 5\}$. If $\deg(v) < 5$, or if the neighbors of v use fewer than 5 colors, we can assign the unused color to v.

Suppose $\deg(v) = 5$ and its neighbors v_1, v_2, v_3, v_4, v_5 (in clockwise order) are colored c_1, c_2, c_3, c_4, c_5 respectively. Let $G_{i,j}$ be the subgraph of G' induced by vertices colored c_i or c_j.

1. If v_1 and v_3 are in different connected components of $G_{1,3}$, we swap colors 1 and 3 in the component containing v_1. Now v_1 is color 3, and color 1 is available for v.
2. If v_1 and v_3 are in the same connected component, there exists a path P between them colored only 1 and 3. This path, along with v, forms a cycle that separates v_2 from v_4.
3. Because the graph is planar, v_2 and v_4 cannot be in the same connected component of $G_{2,4}$. We swap colors 2 and 4 in the component containing v_2. Color 2 is now available for v.

In all cases, G is 5-colorable.

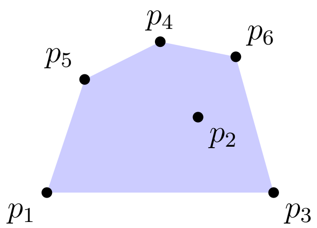
3 Convex Hull

Definition. A set $S \subseteq \mathbb{R}^d$ is **convex** if for any two points $x, y \in S$ and any $\lambda \in [0, 1]$, we have

$$\lambda x + (1 - \lambda) y \in S$$

Definition. The **convex hull** $\text{conv}(S)$ of a set $S \subseteq \mathbb{R}^d$ is the smallest convex set containing S. Equivalently,

$$\text{conv}(S) = \left\{ \sum_{i=1}^{k} \lambda_i x_i \,\middle|\, x_i \in S, \lambda_i \geq 0, \sum_{i=1}^{k} \lambda_i = 1, k \in \mathbb{N} \right\}$$



Theorem. (Carathéodory's Theorem) Let $S \subseteq \mathbb{R}^d$. For every point $x \in \text{conv}(S)$, there exist points $x_1, x_2, \ldots, x_{d+1} \in S$ such that:

$$x \in \text{conv}(\{x_1, x_2, \ldots, x_{d+1}\})$$

That is, every point in the convex hull of S can be expressed as a convex combination of at most $d+1$ points from S.

Proof. Let $x \in \text{conv}(S)$. By definition, there exist points $y_1, y_2, \ldots, y_k \in S$ and coefficients $\lambda_1, \lambda_2, \ldots, \lambda_k \geq 0$ with $\sum_{i=1}^{k} \lambda_i = 1$ such that

$$x = \sum_{i=1}^{k} \lambda_i y_i$$

If $k \leq d+1$, we are done. Assume $k > d+1$. Consider the vectors:

$$v_i = \begin{pmatrix} y_i \\ 1 \end{pmatrix} \in \mathbb{R}^{d+1} \quad \text{for } i = 1, \ldots, k$$

Since $k > d+1$, these vectors are linearly dependent. Hence, there exist $\mu_1, \mu_2, \ldots, \mu_k \in \mathbb{R}$, not all zero, such that

$$\sum_{i=1}^{k} \mu_i v_i = 0 \quad \text{and} \quad \sum_{i=1}^{k} \mu_i = 0$$

Let $\alpha = \min\{\lambda_i/\mu_i : \mu_i > 0\}$. Define new coefficients:

$$\lambda_i' = \lambda_i - \alpha\mu_i \quad \text{for } i = 1, \ldots, k$$

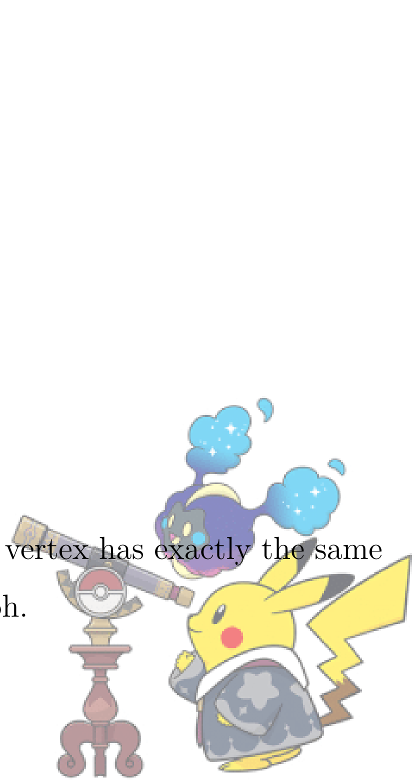
These satisfy

- $\lambda_i' \geq 0$ for all i
- $\sum_{i=1}^{k} \lambda_i' = 1$
- At least one $\lambda_i' = 0$
- $x = \sum_{i=1}^{k} \lambda_i' y_i$

Hence, we have expressed x as a convex combination of at most $k-1$ points.

4 Linear Algebra Method

Definition. (Regular Graph) A graph G is called k-regular if every single vertex has exactly the same number of neighbors, k. This k is known as the degree or valency of the graph.



Definition. **(Special Matrix)** The all-ones matrix, denoted as J, is an $n \times n$ matrix where every single entry is 1.

$$J = \begin{pmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{pmatrix}$$

Definition. **(Adjacency Matrix)** Let $G = (V, E)$ be a graph, where V is the set of vertices and E is the set of edges. The adjacency matrix A of the graph is a square matrix with dimensions $|V| \times |V|$, where each element A_{ij} is defined as

$$A_{ij} = \begin{cases} 1 & \text{if there is an edge between vertex } v_i \text{ and } v_j, \\ 0 & \text{otherwise.} \end{cases}$$

In an undirected graph, the adjacency matrix is symmetric, i.e., $A_{ij} = A_{ji}$, because if there is an edge between v_i and v_j, there is also an edge between v_j and v_i.

Example. For the graph $G = (V, E)$ with vertices $V = \{v_1, v_2, v_3\}$ and edges $E = \{(v_1, v_2), (v_2, v_3)\}$, the adjacency matrix is

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Properties. For $k \geq 1$, $(A^k)_{ij}$ counts the number of **walks of length** k from vertex i to vertex j. There are some special cases:

- **Diagonal entries of** A^2: $(A^2)_{ii}$ = number of walks of length 2 from i to i This equals the **degree** of vertex i, since you must go to a neighbor and back. Therefore, $\text{Tr}(A^2) = \sum_{i=1}^{n} \deg(i) = 2|E|$.
- **Off-diagonal entries of** A^2**:** For $i \neq j$, $(A^2)_{ij} = |N(i) \cap N(j)|$ = number of common neighbors of i and j.
- **Triangles:** $(A^3)_{ii}$ counts closed walks of length 3 from i to i. Each triangle contributes 6 such walks (3 starting vertices $\times$ 2 directions). Hence, $\text{Tr}(A^3) = 6 \times$ (number of triangles).

Definition. A **strongly regular graph** with parameters (n, k, λ, μ) is a k-regular graph on n vertices such that:

1. Any two **adjacent** vertices have exactly λ common neighbors.

2. Any two **non-adjacent** vertices have exactly μ common neighbors.

Proposition. For an SRG(n, k, λ, μ), the adjacency matrix satisfies

$$A^2 = kI + \lambda A + \mu(J - I - A)$$

which simplifies to

$$A^2 = (\lambda - \mu)A + (k - \mu)I + \mu J$$

Proof.

- Diagonal of A^2: k (since regular).
- Off-diagonal, adjacent vertices: λ common neighbors $\Longrightarrow (A^2)_{uv} = \lambda$.
- Off-diagonal, non-adjacent vertices: μ common neighbors $\Longrightarrow (A^2)_{uv} = \mu$.

Proposition. Let θ and τ be the other eigenvalues (besides k). They satisfy

$$\theta + \tau = \lambda - \mu, \quad \theta\tau = \mu - k$$

The multiplicities m_θ and m_τ can be found using:

- $1 + m_\theta + m_\tau = n$
- $\mathrm{Tr}(A) = 0 = k + m_\theta \theta + m_\tau \tau$

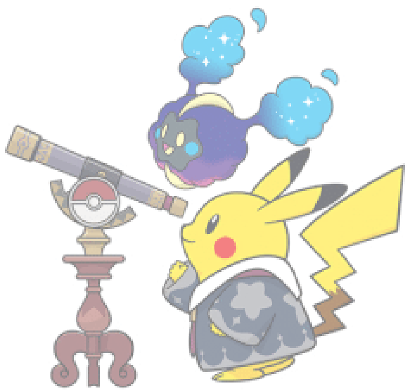
These come from the fact that A has only 3 distinct eigenvalues for connected SRGs (other than complete graphs).

Properties. For a k-regular graph:

- The all-ones vector $\mathbf{1} = (1, 1, \ldots, 1)^T$ is an eigenvector of A with eigenvalue k.
- If G is connected, then k is the **largest eigenvalue** and has multiplicity 1.
- All eigenvalues satisfy $|\lambda| \leq k$.

Example. Let G be a k-regular graph with 10 vertices such that

- No two adjacent vertices have a common neighbor.
- Any two non-adjacent vertices have precisely one common neighbor.



Let A be the adjacency matrix of G.

(a) Show that $A^2 + A - (k-1)I = J$, where I is the identity matrix and J is the all-one matrix.

(b) Determine the value of k.

(c) Find all distinct eigenvalues of A together with their multiplicities.

(d) Construct G and justify your answer rigorously.

Solution.

(a) Let A be the adjacency matrix. The entry $(A^2)_{ij}$ represents the number of walks of length 2 from vertex i to vertex j.

- If $i = j$, $(A^2)_{ii}$ is the degree of vertex i, which is k.
- If i is adjacent to j ($A_{ij} = 1$), the problem states they have 0 common neighbors, so $(A^2)_{ij} = 0$.
- If i is not adjacent to j and $i \neq j$ ($A_{ij} = 0$), they have precisely 1 common neighbor, so $(A^2)_{ij} = 1$.

We can express A^2 as

$$A^2 = kI + 0A + 1(J - I - A)$$

Simplifying the right side:

$$A^2 = kI + J - I - A \implies A^2 = J - A + (k-1)I$$

Rearranging terms gives

$$A^2 + A - (k-1)I = J$$

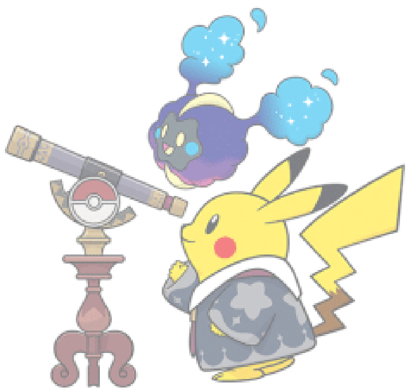
(b) Multiply equation by $\mathbf{1}$:

$$(A^2 + A - (k-1)I)\mathbf{1} = J\mathbf{1}$$

Since $A\mathbf{1} = k\mathbf{1}$, $A^2\mathbf{1} = k^2\mathbf{1}$, $J\mathbf{1} = n\mathbf{1} = 101$:

$$(k^2 + k - (k-1))\mathbf{1} = 10\mathbf{1}$$

$$(k^2 + 1)\mathbf{1} = 10\mathbf{1}$$



$$k^2 + 1 = 10 \implies k^2 = 9 \implies k = 3$$

Hence, G is 3-regular.

(c) With $k = 3$, equation becomes

$$A^2 + A - 2I = J$$

Eigenvector $\mathbf{1}$: $A\mathbf{1} = 3\mathbf{1} \implies J\mathbf{1} = 10\mathbf{1}$. For any eigenvector $v \perp \mathbf{1}$, $Jv = 0$:

$$A^2v + Av - 2v = 0$$

If $Av = \lambda v$, then $\lambda^2 + \lambda - 2 = 0 \implies (\lambda + 2)(\lambda - 1) = 0$. So other eigenvalues are -2 and 1. Let multiplicities be $1, m_1, m_2$ for $3, 1, -2$ respectively, with $m_1 + m_2 = 9$. Trace condition gives $\mathrm{Tr}(A) = 0 \implies 3 + m_1(1) + m_2(-2) = 0$. Solving $3 + m_1 - 2m_2 = 0$ and $m_1 = 9 - m_2$:

$$m_2 = 4,\ m_1 = 5.$$

Therefore, the answer is $3^{(1)}, 1^{(5)}, -2^{(4)}$.

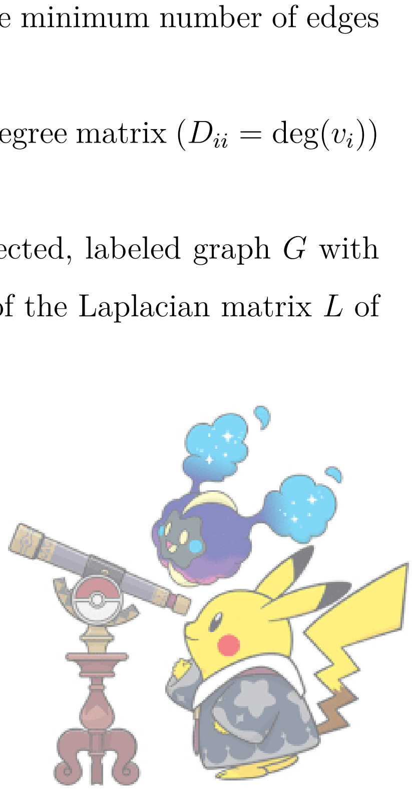
(d) Construct a graph with vertices of 2-element subsets of $\{1, 2, 3, 4, 5\}$, edges between disjoint subsets, as it is 3-regular, there is no triangles ($\lambda = 0$), and any two non-adjacent vertices share exactly one common neighbor ($\mu = 1$).

Definition. A **spanning tree** of a graph G is a subgraph that includes all the vertices of G and is a tree. This means that a spanning tree connects all the vertices of the graph with the minimum number of edges and without forming any cycles.

Definition. The Laplacian is defined as $L = D - A$, where D is the diagonal degree matrix ($D_{ii} = \deg(v_i)$) and A is the adjacency matrix.

Theorem. (Kirchhoff's Matrix Tree Theorem) For a connected, undirected, labeled graph G with n vertices, the total number of spanning trees $\tau(G)$ is given by any cofactor of the Laplacian matrix L of G, where

$$L_{ij} = \begin{cases} \deg(v_i) & \text{if } i = j \\ -1 & \text{if } i \neq j \text{ and } v_i \sim v_j \\ 0 & \text{otherwise} \end{cases}$$



that is, delete any row r and column c from L, and

$$\tau(G) = \det\left(L^{(r,c)}\right)$$

where $L^{(r,c)}$ is the $(n-1)\times(n-1)$ matrix with row r and column c removed.

Example. State the Matrix-Tree Theorem for counting the total number of different spanning trees in a labeled graph. Find this number for the complete bipartite graph $K_{m,n}$.

Solution. The complete bipartite graph $K_{m,n}$ has vertex set partitioned into m vertices $u_1,\dots,u_m$ and n vertices $v_1,\dots,v_n$, with all edges between the two parts. Ordering vertices as $u_1,\dots,u_m,v_1,\dots,v_n$, the Laplacian is

$$L = \begin{pmatrix} nI_m & -\mathbf{1}_{m\times n} \\ -\mathbf{1}_{n\times m} & mI_n \end{pmatrix}$$

where $\mathbf{1}_{p\times q}$ denotes the $p\times q$ all-ones matrix. The eigenvalues of $L(K_{m,n})$ are

- 0 (multiplicity 1)
- m (multiplicity $n-1$)
- n (multiplicity $m-1$)
- $m+n$ (multiplicity 1)

provided $m,n\geq 1$. The product of the non-zero eigenvalues is

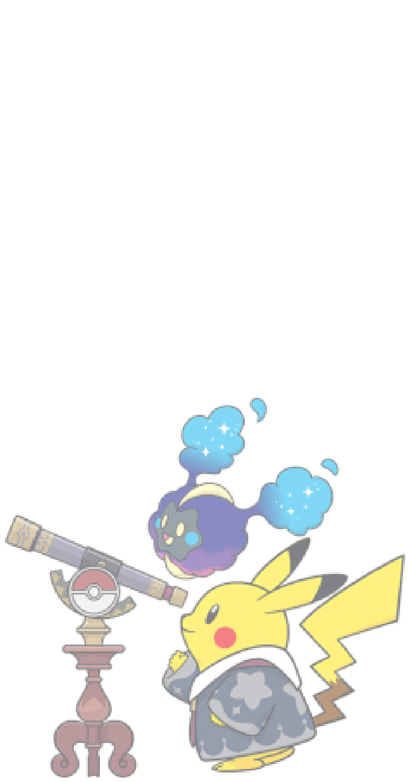
$$m^{n-1}\cdot n^{m-1}\cdot(m+n)$$

By Kirchhoff's theorem, the number of spanning trees is

$$\tau(K_{m,n}) = \frac{1}{m+n}\left[m^{n-1}\cdot n^{m-1}\cdot(m+n)\right]$$

Hence,

$$\tau(K_{m,n}) = m^{n-1}\,n^{m-1}$$



5 Probabilistic Method

Theorem. (Boole's Inequality or Union Bound) Boole's inequality states that for any finite or countable set of events $A_1, A_2, A_3, \ldots,$

$$P\left(\bigcup_i A_i\right) \leq \sum_i P(A_i)$$

Proof. Consider the case where there are only two events A_1 and A_2. We want to show that

$$P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$$

Using the inclusion-exclusion principle, we have:

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2)$$

Since $P(A_1 \cap A_2) \geq 0$, it follows that

$$P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$$

Hence, the base case holds. Assume that the inequality holds for any collection of n events, i.e.,

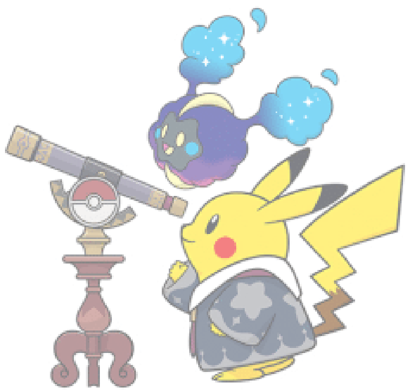
$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$$

We now show that the inequality holds for $n+1$ events. Consider the events $A_1, A_2, \ldots, A_n, A_{n+1}$. Using the inclusion-exclusion principle, we write:

$$P\left(\bigcup_{i=1}^{n+1} A_i\right) = P\left(\bigcup_{i=1}^{n} A_i \cup A_{n+1}\right)$$

Similar to the above case,

$$P\left(\bigcup_{i=1}^{n} A_i \cup A_{n+1}\right) \leq P\left(\bigcup_{i=1}^{n} A_i\right) + P(A_{n+1})$$



By the inductive hypothesis,

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$$

Hence,

$$P\left(\bigcup_{i=1}^{n+1} A_i\right) \leq \sum_{i=1}^{n} P(A_i) + P(A_{n+1}) = \sum_{i=1}^{n+1} P(A_i)$$

Therefore, by induction, Boole's Inequality holds for any finite or countable set of events.

Example. $R(k,k) > 2^{k/2}$ for $k \geq 3$

Solution. Consider a complete graph K_n. We color each of its $\binom{n}{2}$ edges either red or blue independently and uniformly at random (each color has a probability of $1/2$). The sample space (Ω) is given that the set of all $2^{\binom{n}{2}}$ possible 2-colorings of K_n. Probability measure is that each specific coloring is equally likely. A coloring is bad if it contains at least one monochromatic K_k. For every subset of vertices $S \subseteq V$ such that $|S| = k$, let A_S be the event that the subgraph induced by S is monochromatic. There are $\binom{k}{2}$ edges in a K_k. The probability that all these edges are red is $\left(\frac{1}{2}\right)^{\binom{k}{2}}$. The probability that all these edges are blue is $\left(\frac{1}{2}\right)^{\binom{k}{2}}$. Since these two outcomes are disjoint,

$$P(A_S) = \left(\frac{1}{2}\right)^{\binom{k}{2}} + \left(\frac{1}{2}\right)^{\binom{k}{2}} = 2^{1-\binom{k}{2}}$$

We want to show that the probability that any bad event occurs is less than 1. There are $\binom{n}{k}$ such subsets S. By the Union Bound,

$$P(\text{any } A_S \text{ occurs}) = P\left(\bigcup_{|S|=k} A_S\right) \leq \sum_{|S|=k} P(A_S) = \binom{n}{k} 2^{1-\binom{k}{2}}$$

If this sum is strictly less than 1, then there must be a coloring in our space where no bad event A_S occurs. We use the inequality $\binom{n}{k} < \frac{n^k}{k!}$. We want:

$$\frac{n^k}{k!} 2^{1-\frac{k(k-1)}{2}} < 1$$

Note that when $n \leq 2^{k/2}$,

$$\frac{n^k}{k!} 2^{1-\frac{k(k-1)}{2}} \leq \frac{(2^{k/2})^k}{k!} 2^{1-\frac{k^2-k}{2}} = \frac{2^{k^2/2}}{k!} 2^{1-\frac{k^2}{2}+\frac{k}{2}} = \frac{2^{1+k/2}}{k!} < 1$$

for $k \geq 3$. There exists at least one coloring where no subgraph of order k is monochromatic. Therefore,

$$R(k,k) > 2^{k/2} \text{ for } k \geq 3$$

Definition. A **k-subset** of a set S is a subset that contains exactly k elements of S.

Example. Consider the family $\mathcal{F}$ of all ordered pairs (A, B) of disjoint k-subsets of $[n] = \{1, 2, \ldots, n\}$. A subset X of $[n]$ is said to *separate* the pair (A, B) if $A \subseteq X$ and $B \cap X = \emptyset$. Prove that if

$$\binom{n}{k}\binom{n-k}{k}(1 - 2^{-2k})^m < 1$$

for some positive integer m, then there exist m subsets $X_1, X_2, \ldots, X_m$ of $[n]$ such that every pair (A, B) in $\mathcal{F}$ is separated by at least one of them.

Solution. We use the probabilistic method. Let $X_1, X_2, \ldots, X_m$ be m subsets of $[n]$ chosen independently and uniformly at random from the power set $\mathcal{P}([n])$. This is equivalent to saying that for each X_i and each element $j \in [n]$, the probability $P(j \in X_i) = 1/2$, independent of all other elements and other subsets. First, let us calculate the probability that a single random subset X separates a fixed pair $(A, B) \in \mathcal{F}$. For X to separate (A, B), we require

1. $j \in X$ for all $j \in A$ (which happens with probability $(1/2)^{|A|} = 2^{-k}$)

2. $j \notin X$ for all $j \in B$ (which happens with probability $(1/2)^{|B|} = 2^{-k}$)

Since A and B are disjoint, these $2k$ events are independent. Hence,

$$P(X \text{ separates } (A, B)) = 2^{-k} \cdot 2^{-k} = 2^{-2k}$$

The probability that X *does not* separate (A, B) is $1 - 2^{-2k}$. Since the m subsets $X_1, \ldots, X_m$ are chosen independently, the probability that **none** of them separate (A, B) is

$$P(\text{no } X_i \text{ separates } (A, B)) = (1 - 2^{-2k})^m$$

Now, let E be the event that there exists at least one pair $(A, B) \in \mathcal{F}$ that is not separated by any X_i. By the Union Bound:

$$P(E) = P\left(\bigcup_{(A,B)\in\mathcal{F}} \{\text{no } X_i \text{ separates } (A, B)\}\right) \leq \sum_{(A,B)\in\mathcal{F}} P(\text{no } X_i \text{ separates } (A, B))$$

The number of such pairs (A, B) in $\mathcal{F}$ is determined by choosing k elements for A from n, and then k elements for B from the remaining $n-k$. Thus, $|\mathcal{F}| = \binom{n}{k}\binom{n-k}{k}$. Substituting this into the inequality:

$$P(E) \leq \binom{n}{k}\binom{n-k}{k}(1 - 2^{-2k})^m$$

According to the given condition, this expression is strictly less than 1. Therefore, $P(E) < 1$. This implies that the probability of the complement event (that *all* pairs are separated by at least one X_i) is strictly greater than 0:

$$P(E^c) = 1 - P(E) > 0$$

Since the probability is non-zero, there must exist at least one specific collection of m subsets $\{X_1, \dots, X_m\}$ that separates every pair in $\mathcal{F}$.

Theorem. (Linearity of Expectation) If $X_1, X_2, \dots, X_n$ are random variables, then

$$\mathbb{E}[X_1 + X_2 + \cdots + X_n] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n]$$

Theorem. (Markov's Inequality) Let X be a non-negative random variable ($X \geq 0$) with an expected value $E[X]$. For any positive constant $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof. The expectation of X is calculated by integrating over all possible values:

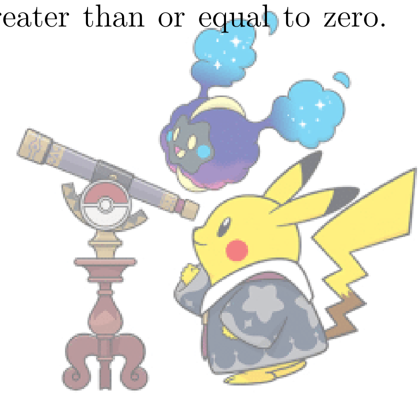
$$E[X] = \int_0^\infty x f(x) dx$$

We can break the integral into two parts: the interval from 0 to a, and the interval from a to ∞.

$$E[X] = \int_0^a x f(x) dx + \int_a^\infty x f(x) dx$$

Since X is non-negative ($X \geq 0$), the first integral $\int_0^a x f(x) dx$ must be greater than or equal to zero. Therefore, if we drop it, we get

$$E[X] \geq \int_a^\infty x f(x) dx$$



For all values of x in the interval $[a, \infty)$, we know that $x \geq a$. If we replace x with a in the integral, the value will only get smaller or stay the same:

$$E[X] \geq \int_a^\infty af(x)dx$$

Since a is a constant, we can move it outside the integral:

$$E[X] \geq a \int_a^\infty f(x)dx$$

The integral $\displaystyle\int_a^\infty f(x)dx$ is, by definition, the probability that $X \geq a$:

$$E[X] \geq a \cdot P(X \geq a)$$

which gives the Markov's inequality.

Theorem. (Chebyshev's Inequality) It states that for any random variable X with expected value $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X)$,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Proof. We begin by applying the definition of variance:

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

Now, let's consider the event $|X - \mu| \geq k\sigma$. Squaring both sides of the inequality, we get

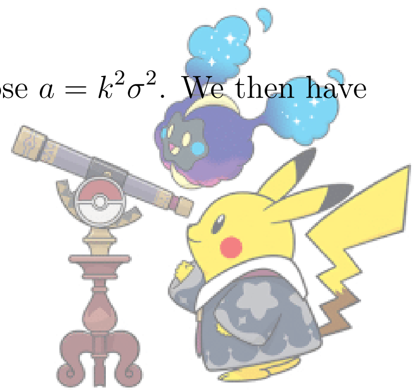
$$(X - \mu)^2 \geq k^2\sigma^2$$

Hence, the probability of the event $|X - \mu| \geq k\sigma$ is the same as the probability that $(X - \mu)^2 \geq k^2\sigma^2$, i.e.

$$P(|X - \mu| \geq k\sigma) = P((X - \mu)^2 \geq k^2\sigma^2)$$

We can apply Markov's inequality to the random variable $(X - \mu)^2$ and choose $a = k^2\sigma^2$. We then have

$$P((X - \mu)^2 \geq k^2\sigma^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2\sigma^2}$$



Since $\mathbb{E}[(X-\mu)^2] = \text{Var}(X) = \sigma^2$, this becomes

$$P((X-\mu)^2 \geq k^2\sigma^2) \leq \frac{\sigma^2}{k^2\sigma^2}$$

Simplifying the right-hand side, we get

$$P((X-\mu)^2 \geq k^2\sigma^2) \leq \frac{1}{k^2}.$$

Therefore, we have

$$P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Proposition. (Deletion Method) Let G be a random graph or other combinatorial structure, and let X be a random variable representing the number of edges or other elements that satisfy a given property. The Deletion Method proceeds as follows:

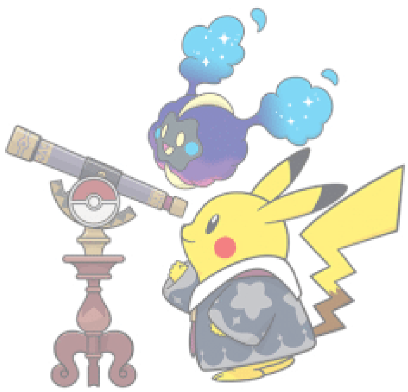
1. Start with a structure G (for example, a random graph or a set of points).
2. Randomly delete a subset of elements from G, where each element is deleted independently with some probability p.
3. Let X' be the random variable that counts the number of elements of interest remaining after deletion. Show that $\mathbb{E}[X']$ is large enough to guarantee that there exists a remaining structure with the desired properties.
4. Use Markov's inequality to bound the probability that X' is large enough to ensure the existence of the desired structure.

Theorem. (Lovász Local Lemma) Let $A_1, A_2, \ldots, A_n$ be a finite set of events in a probability space. Suppose that each event A_i is independent of all but at most d other events A_j. If for all i,

$$P(A_i) \leq p$$

and the following condition holds:

$$ep(d+1) \leq 1$$



(where $e \approx 2.718$ is the base of the natural logarithm), then

$$P\left(\bigcap_{i=1}^{n} \overline{A_i}\right) > 0$$

Proof. We define a dependency graph $G = (V, E)$ where the vertices $V = \{1, \ldots, n\}$ represent our events. An edge exists between i and j if A_i and A_j are dependent. According to the problem, the maximum degree of this graph is $\Delta(G) \leq d$. For any event A_i, let $\Gamma(i)$ denote the set of its neighbors in G (the events it depends on). By definition, $|\Gamma(i)| \leq d$. A_i is mutually independent of all events in $V \setminus (\{i\} \cup \Gamma(i))$. To show $P\left(\bigcap_{i=1}^{n} \overline{A_i}\right) > 0$, it suffices to prove that for any i and any subset $S \subset \{1, \ldots, n\}$ that does not contain i:

$$P\left(A_i \middle| \bigcap_{j \in S} \overline{A_j}\right) \leq \frac{1}{d+1}$$

If this holds, then by the chain rule of probability:

$$P\left(\bigcap_{i=1}^{n} \overline{A_i}\right) = \prod_{i=1}^{n} P\left(\overline{A_i} \middle| \bigcap_{j<i} \overline{A_j}\right) = \prod_{i=1}^{n} \left(1 - P\left(A_i \middle| \bigcap_{j<i} \overline{A_j}\right)\right)$$

Since each term in the product is at least $1 - \frac{1}{d+1} > 0$, the entire product is strictly greater than zero. We proceed by induction on the size of the set S, $|S| = k$.

Base Case ($k = 0$): When $S = \emptyset$, we simply have $P(A_i) \leq p$. Given the condition $ep(d+1) \leq 1$, and since $e > 1$, we have:

$$p \leq \frac{1}{e(d+1)} < \frac{1}{d+1}$$

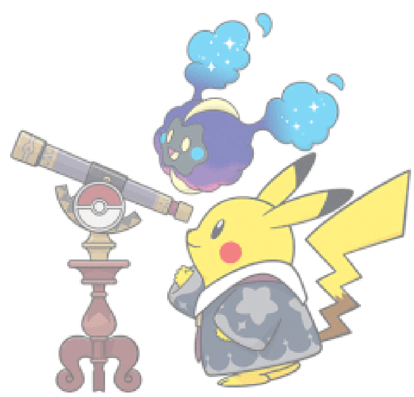
The base case holds.

Inductive Step: Assume the claim holds for all sets S' with $|S'| < k$. We want to prove it for $|S| = k$. Split S into two parts: First, $S_1 = S \cap \Gamma(i)$ (the neighbors of A_i). Note $|S_1| \leq d$. Second, $S_2 = S \setminus S_1$ (the events independent of A_i). Using the definition of conditional probability:

$$P\left(A_i \middle| \bigcap_{j \in S} \overline{A_j}\right) = \frac{P(A_i \cap \bigcap_{j \in S_1} \overline{A_j} \mid \bigcap_{j \in S_2} \overline{A_j})}{P(\bigcap_{j \in S_1} \overline{A_j} \mid \bigcap_{j \in S_2} \overline{A_j})}$$

As A_i is independent of all events in S_2,

$$\text{Numerator} \leq P(A_i \mid \bigcap_{j \in S_2} \overline{A_j}) = P(A_i) \leq p$$



Let $S_1 = \{j_1, j_2, \dots, j_m\}$ where $m \le d$. We expand the denominator using the chain rule:

$$P(\overline{A_{j_1}} \cap \dots \cap \overline{A_{j_m}} \mid \bigcap_{j \in S_2} \overline{A_j}) = \prod_{r=1}^{m} P(\overline{A_{j_r}} \mid \overline{A_{j_1}} \cap \dots \cap \overline{A_{j_{r-1}}} \cap \bigcap_{j \in S_2} \overline{A_j})$$

Each term in this product is of the form $P(\overline{A_{j_r}} \mid \bigcap_{k \in S'} \overline{A_k})$ where $|S'| < k$. By our inductive hypothesis,

$$P(A_{j_r} \mid \dots) \le \frac{1}{d+1} \implies P(\overline{A_{j_r}} \mid \dots) \ge 1 - \frac{1}{d+1}$$

Hence, the denominator is bounded by

$$\text{Denominator} \ge \left(1 - \frac{1}{d+1}\right)^m \ge \left(1 - \frac{1}{d+1}\right)^d$$

Combining the numerator and denominator, we have

$$P\left(A_i \middle| \bigcap_{j \in S} \overline{A_j}\right) \le \frac{p}{(1 - \frac{1}{d+1})^d}$$

To complete the induction, we need this to be $\le \dfrac{1}{d+1}$. This is equivalent to

$$p \le \frac{1}{d+1}\left(1 - \frac{1}{d+1}\right)^d$$

Note that $(1 - \dfrac{1}{d+1})^d > \dfrac{1}{e}$ for $d \ge 1$. The given condition $ep(d+1) \le 1$ can be rewritten as $p \le \dfrac{1}{e(d+1)}$. Altogether,

$$P(A_i \mid \bigcap_{j \in S} \overline{A_j}) \le \frac{1}{e(d+1)} \cdot e = \frac{1}{d+1}$$

The induction is complete, and the result is proved.